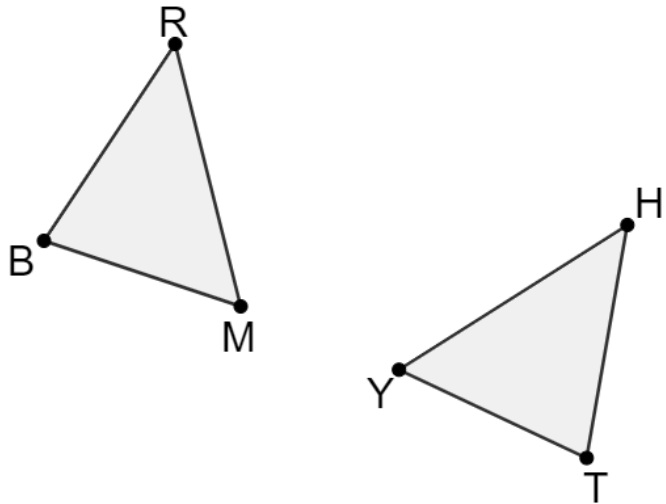


Triangles BRM and YHT are shown. Determine if the given information is sufficient to conclude that the triangles are congruent. If so, name the congruence theorem or postulate that shows congruence.



Given Information	Are They Congruent?	Theorem or Postulate Used
1. $\overline{MR} \cong \overline{YH}$, $\angle HYT \cong \angle BMR$, $\angle BRM \cong \angle THY$		
2. $\overline{MB} \cong \overline{YT}$, $\overline{MR} \cong \overline{HY}$, $\overline{BR} \cong \overline{TH}$		
3. $\overline{MB} \cong \overline{YT}$, $\angle YTH \cong \angle MBR$, $\angle BRM \cong \angle THY$		

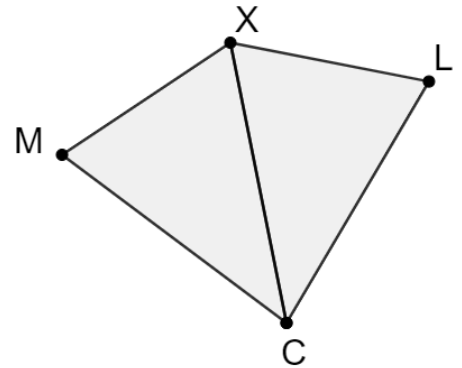
4. Quadrilateral $CLXM$ is shown where $\overline{MC} \cong \overline{LC}$ and $\angle MCX \cong \angle XCL$.
Complete the paragraph proof.

It is given that $\overline{MC} \cong \overline{LC}$ and $\angle MCX \cong \angle XCL$.

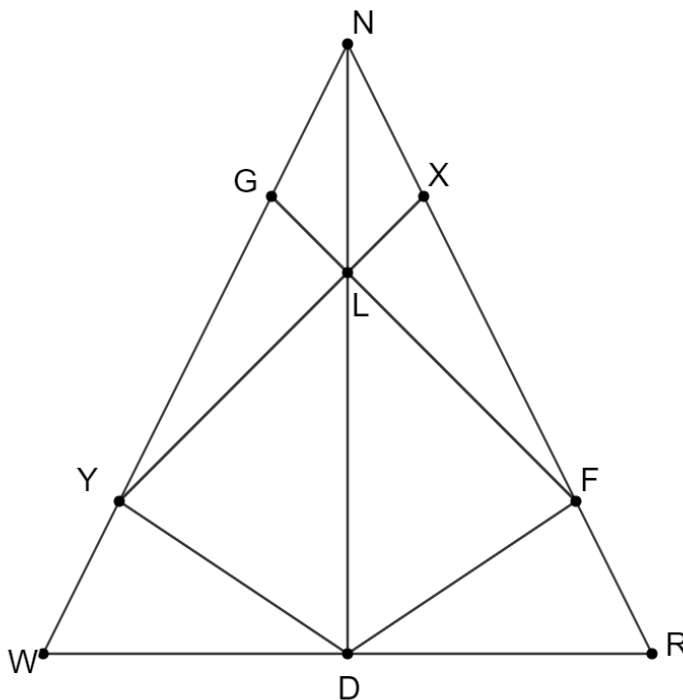
By the reflexive property, _____.

Therefore, Δ _____ $\cong \Delta$ _____

by _____.



5. Triangle NRW is shown.

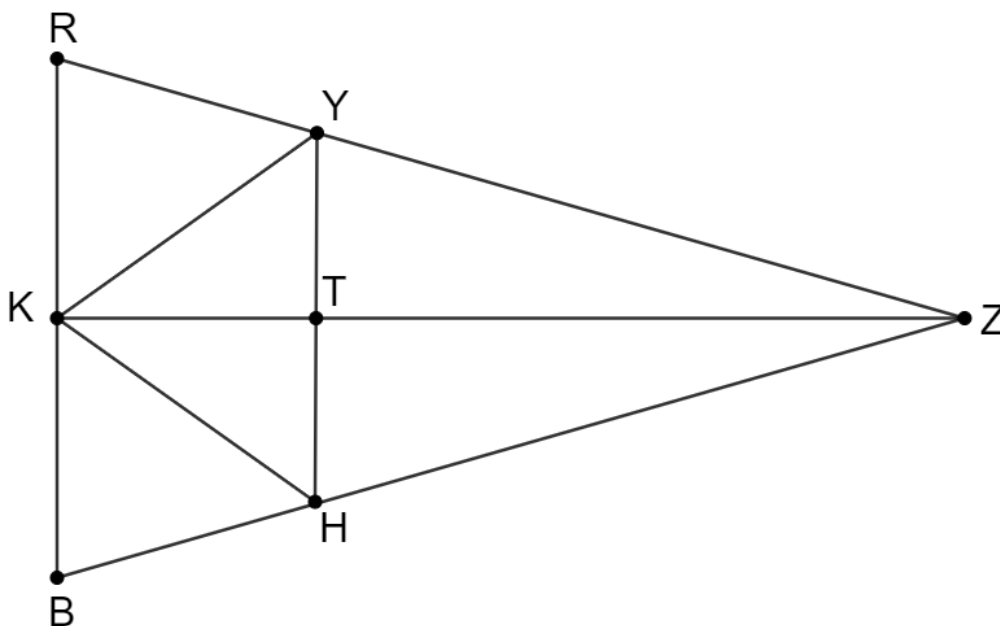


Complete each statement.

- a. If $\triangle FLD \cong \triangle YLD$ then $\angle DFL \cong \angle$ _____.

- b. If $\overline{ND} \perp \overline{WR}$ and $\overline{ND} \cong \overline{ND}$, the information needed to show that
 $\triangle NDW \cong \triangle NDR$ using HL is _____.

6. Triangle RZB is shown.



Given: $\triangle KYT \cong \triangle KHT$

Prove: $\triangle YKZ \cong \triangle HKZ$

An incomplete proof is shown.

Statement	Reason
1. $\triangle KYT \cong \triangle KHT$	1. Given
2. $\overline{KY} \cong \overline{KH}$	2.
3.	3. Corresponding Parts of Congruent Triangles are Congruent (CPCTC)
4.	4. Reflexive Property
5. $\triangle YKZ \cong \triangle HKZ$	5.

What is the reason for step 5?

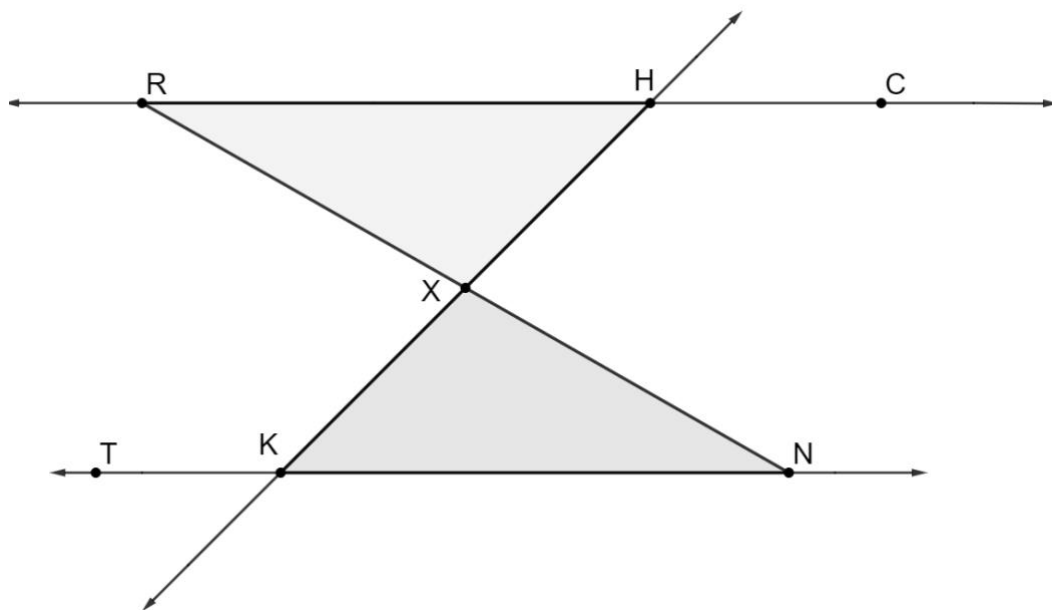
Reason for step 5	
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7. Parallel lines $\overleftrightarrow{RC}$ and $\overleftrightarrow{TN}$ with transversal $\overleftrightarrow{KH}$ is shown.

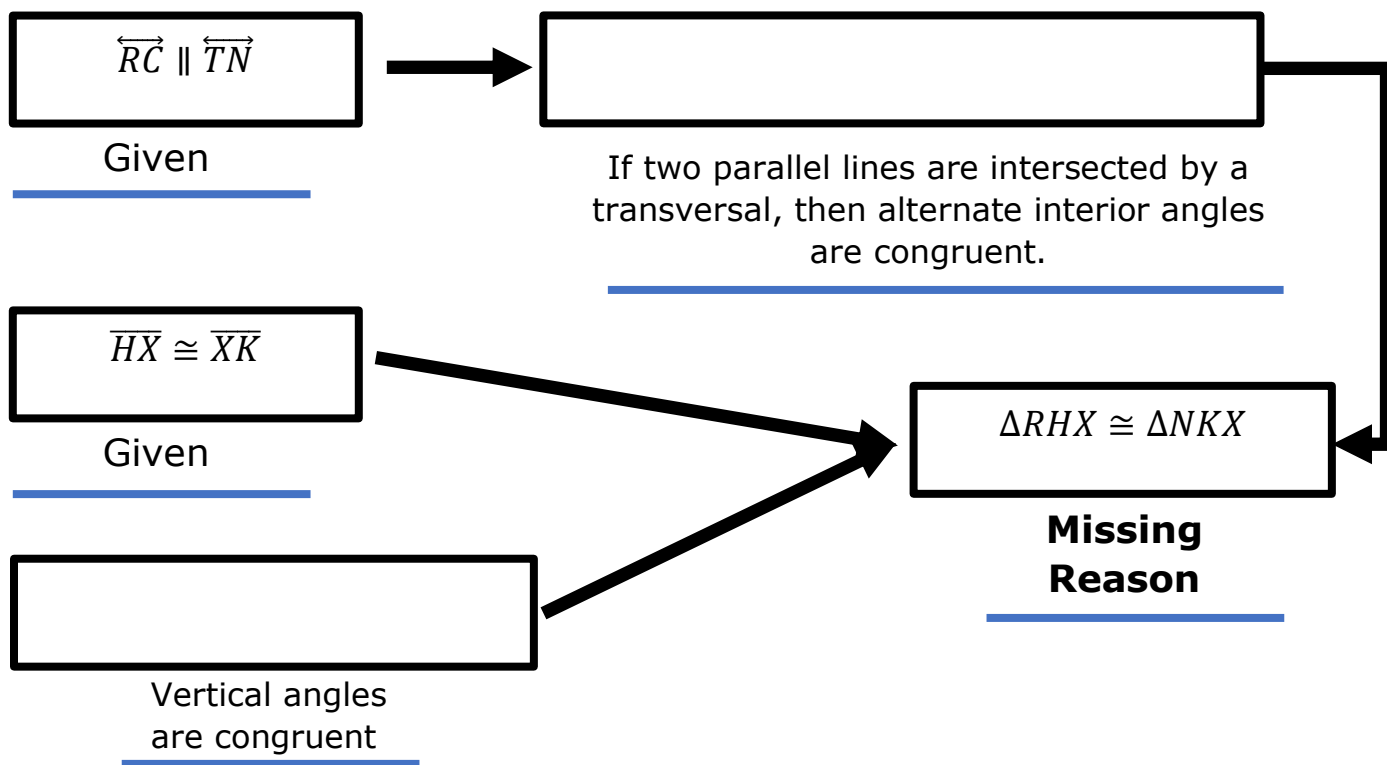
Given: $\overleftrightarrow{RC} \parallel \overleftrightarrow{TN}$,

$\overline{HX} \cong \overline{KX}$

Prove: $\triangle RHX \cong \triangle NKX$



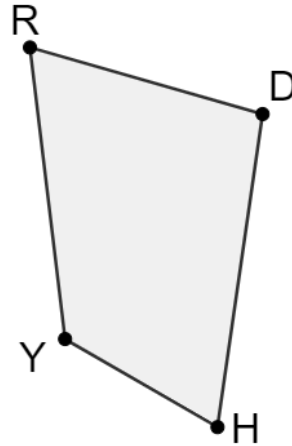
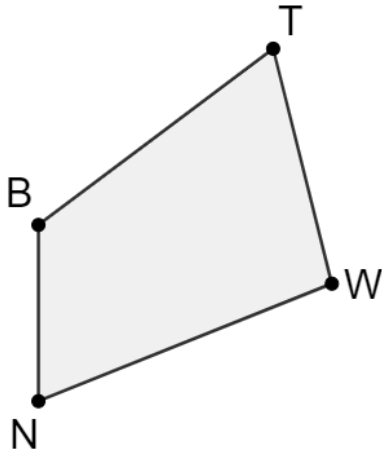
An incomplete flowchart proof is shown.



What is the missing reason in the flowchart proof?

Missing Reason	
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8. Quadrilaterals $TWNB$ and $HYRD$ are shown.



It is given that $TWNB \cong RDHY$, $NW = 3x$, $RD = 5x - 19.6$, $RY = 2x + 5.6$, $HD = 5x - 14.4$, $TW = 3x - 5.2$, and $BN = x + 4.8$.

- a. Determine the value of x . Show your work.

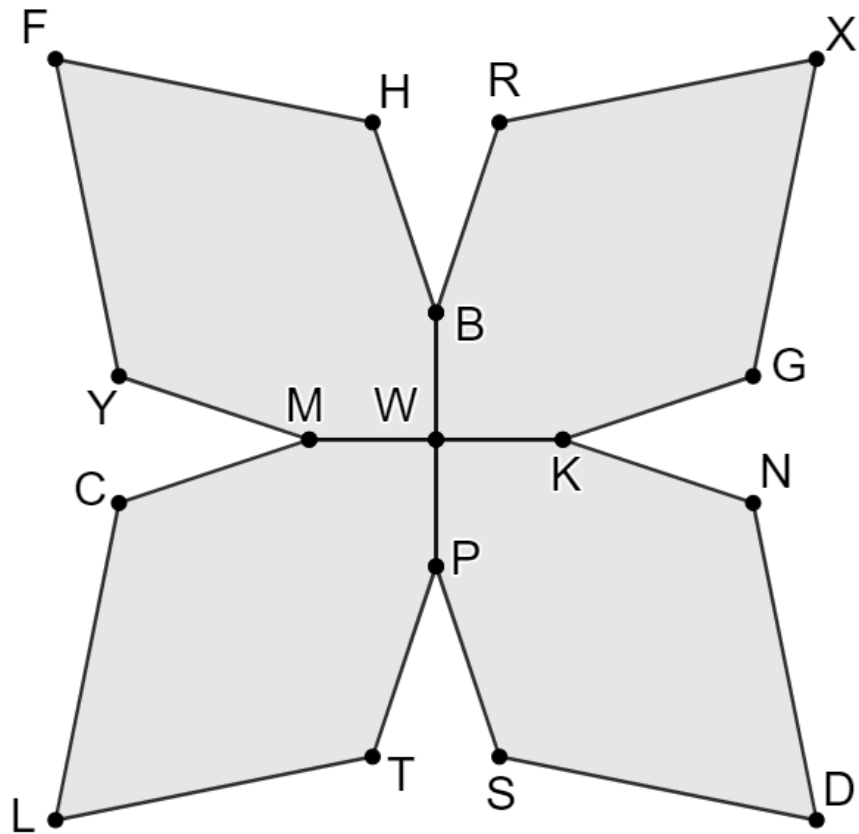
$x = \underline{\hspace{2cm}}$

- b. Determine the perimeter of quadrilateral $TWNB$. Show your work.

Perimeter	
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9. A graphic artist created a design for a flower using four congruent hexagons as shown.

Given: $\overline{BP} \perp \overline{MK}$,
 $m\angle MYF = m\angle FHB$,
 $m\angle WMC = m\angle WPT$,
 $m\angle GXR = 66^\circ$,
 $m\angle DNK = (3x + 18)^\circ$, and
 $m\angle WBR = (5x - 8)^\circ$. The sum of the angles in $WKNDSP$ is 720° .



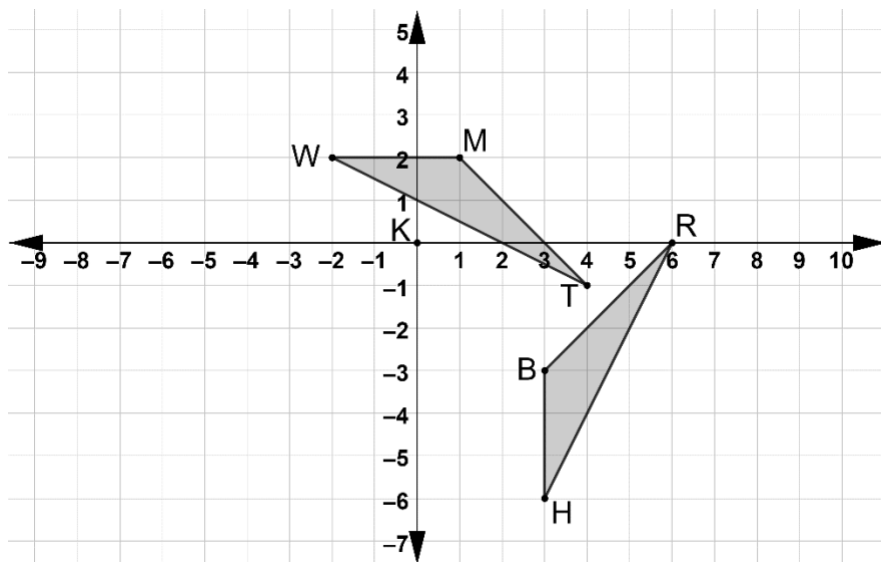
Find each. Show your work

a. $m\angle TLC =$ _____

b. $m\angle KGX =$ _____

Work

10. Triangles TMW and BHR are shown, where $\overline{HB} \cong \overline{WM}$, $\angle WMT \cong \angle HBR$, and $\angle HRB \cong \angle WTM$.



- a. Describe the transformations that will map $\triangle RBH$ to $\triangle TMW$.

- b. Complete the statements.

Because the mapping described consists of

- ☐ dilations,
- ☐ rigid motions,
- ☐ both dilations and rigid motions,

the definition of

- ☐ similarity
- ☐ congruence

can be applied to show that

- ☐ the vertices
- ☐ corresponding sides
- ☐ corresponding angles
- ☐ both corresponding sides and corresponding angles

are

- ☐ similar.
- ☐ congruent.

Therefore, since $\overline{HB} \cong \overline{WM}$, $\angle WMT \cong \angle HBR$, and $\angle HRB \cong \angle WTM$ were

given, then

- ☐ AAS
- ☐ ASA
- ☐ SAS
- ☐ SSS

is sufficient to show triangles are

- ☐ similar.
- ☐ congruent.